

DOWE FLATS PROJECT SUMMARY

MINING, RECLAMATION AND WATER STORAGE

**Southdown, Inc.
(Southwestern Portland Cement)**

May 3, 1993

THE PROPONENT (Permit Applicant)

Southdown, Inc., which does business in Colorado under the name Southwestern Portland Cement (Southwestern), owns and operates a cement plant and related limestone quarries in unincorporated Boulder County near Lyons, Colorado. They are seeking approval for limestone mining near the plant in an area called Dowe Flats, which also is in unincorporated Boulder County. A dam and reservoir is also being proposed in Dowe Flats as a companion land use to mining and reclamation, and potentially as the permanent, post-mining land use.

The cement plant was built in 1969 and was purchased by Southwestern from Martin Marietta Cement in 1984. Southwestern owns or controls approximately 3,375 acres in this area and, accordingly, their future land use plans are an important part of comprehensive planning for this part of northern Boulder County. The vicinity map included in this summary shows the land owned or controlled by Southwestern. The map in the pocket of this summary shows the location of Dowe Flats in relation to Lyons, Longmont and the surrounding area.

Southwestern employs 90 men and women and has an annual payroll of \$3.5 million. The company helps support the local economy, not only with jobs, but through the purchase of over \$10.5 million worth of goods and services each year. Also, Southwestern pays approximately \$850,000 in property taxes annually, while placing less of a demand on government services than residential development. Southwestern's payroll, purchasing and property tax dollars multiply as they pass through the local and regional economy to the point where the cement plant contributes about \$30 million annually to the prosperity of the area.

MINING AND RECLAMATION PLANS

Limestone and shale are mined and processed at the plant to make cement. Limestone mining currently takes place on the cement plant property and at a quarry in Larimer County. About 800,000 tons of limestone and shale are needed every year for cement manufacture. Mining at Dowe Flats would eventually replace the other mining operations.

Some facts about the proposed Dowe Flats limestone mine are:

- Mining and reclamation is proposed for 25 years.
- Annual production is 760,000 tons.
- The deepest part of the mine will be about 100 feet.
- 15 million cubic yards of soil and rock will be used to reclaim the land and create the final land surface.

Reclamation will start immediately and will take place concurrently with mining for the life of the operation. The reclamation goals are to enhance both the esthetic and ecosystem values of the valley by creating hills, valleys, ponds, wetlands, cliffs and grasslands. Diverse topography,

vegetation and water features will greatly improve wildlife habitat over what is now there.

Mining will involve the removal and salvage of topsoil, excavation and blasting of overburden (overburden is rock and soil material overlying the limestone which must be removed in order to mine the limestone) and blasting and excavation of the limestone. The mine will advance from south to north at the rate of about 300 feet per year. Topsoil and overburden will be placed in Dowe Flats according to the reclamation plan to start creating the final land surface. This will be accomplished by scrapers, dozers, and haul trucks. Limestone will be loaded into 25 ton dump trucks and hauled to the cement plant.

The haul trucks will leave the mining property onto County Road 47, then turn on Colorado Highway 66 (Ute Highway) and then turn into County Road 49 (the entrance to the cement plant). This is a total distance of 4,450 feet from the mining property to the cement plant property. All necessary permits will be obtained and all required improvements to public roads will be performed for the hauling on public roads.

County Road 47 goes through the mining area and it is proposed to be relocated to insure public safety and continued access. The proposed relocation is shown on the vicinity map.

Two computer designed color photographs are included in this summary to show the existing valley and the concept of the final reclamation plan.

WATER STORAGE PLANS

Dowe Flats has long been recognized as a potential reservoir site. Southwestern wishes to identify and preserve this off-channel water storage site by obtaining zoning approval for a dam and reservoir. Southwestern does not propose to construct the reservoir at this time and the eventual builder of the reservoir and user(s) of the water are currently unknown. The mining and reclamation plans have been designed to fully accommodate dam construction at any time during or after mining. In fact, the mining and reclamation plans add more water storage capacity (because of the mine pits) and make dam construction material available at low cost. Water studies indicate that a reservoir at Dowe Flats may become viable in about 25 years. It is important to note that Southwestern will guarantee (and post a substantial bond) that mining and reclamation will happen as planned. However, Southwestern can not guarantee that a reservoir will ever be built. When and if dam construction occurs, additional County and State permits will have to be obtained, and this process will include public review and comment opportunities.

A computer designed color photograph is included in this report to show how a reservoir might look after mining and reclamation was completed.

ENVIRONMENTAL, LAND USE, AND COMMUNITY ISSUES

Southwestern recognizes that the economic development associated with cement manufacture and water storage must be balanced with environmental, land use, and community values. Southwestern is an integral part of the community, and their land ownership and land use is, in large part, responsible for the open, rural character of the area. The proposed plans will maintain that character and will be compatible with existing and zoned land uses in the vicinity.

Extensive environmental and land use studies have been and are being performed. These include:

- air quality analysis, including an on-site meteorological station and over two years of air quality baseline data,
- surface and groundwater quality studies and monitoring, including one year of water quality monitoring,
- vegetation and soils studies, sampling and analysis,
- threatened and endangered species studies,
- over 200 drill holes with associated sampling and testing of the soil and bedrock,
- wetlands mapping and analysis,
- mine engineering and reclamation planning studies and analysis,
- detailed wildlife analysis and mapping,
- extensive archaeologic and historic surveys, reports and sampling,
- geotechnical and engineering studies,
- noise analysis, and
- transportation planning and traffic impact analysis.

All of this data is available for public review. Southwestern has assembled a highly qualified team of expert consultants, many of whom live in Boulder County and are personally committed to maintaining the quality of the local environment.

An extensive public and government agency review process is planned whereby all comments

and concerns will be made available to Southwestern. After Southwestern reviews public comments, detailed mitigation plans will be developed to eliminate or minimize project impacts. Necessary or desired modifications to the project will also be carefully evaluated at this time.

PERMITS/APPROVALS REQUIRED

Mining & Reclamation

Boulder County

- Special Use Permit
- Site Specific Development Plan
- Development Agreement/Vesting
- Vacate Existing County Rd. 47
- Approve Relocated County Rd. 47
- Building & Grading Permits

Colorado

- Air Quality Permit
- Mining and Reclamation Permit
- State Highway Access Permit

Federal

- U.S. Army Corp of Engineers
Nationwide Permit

Reservoir

Boulder County

- Special Use Permit

Colorado

- None (presently)

Federal

- U.S. Army Corp of Engineers
Nationwide Permit

PUBLIC REVIEW PROCESS

Southwestern is proposing an extensive public review process in order to give all interested parties adequate time, information and opportunity to provide input into the project.

The main features of the public review process are an extended County referral time and an on-going community information program sponsored by Southwestern. The community information program has started and will continue throughout the permit process. Community and agency review will generate comments over a period of time (probably 3-4 months). After Southwestern has received the comments and has had time to synthesize them, impact mitigation plans will be finalized, project modifications will be made (if appropriate), and additional data collection and analyses will be performed if necessary. The flow chart included in this report shows the public and agency review process.

PROJECT TEAM / CONTACT PEOPLE

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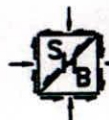
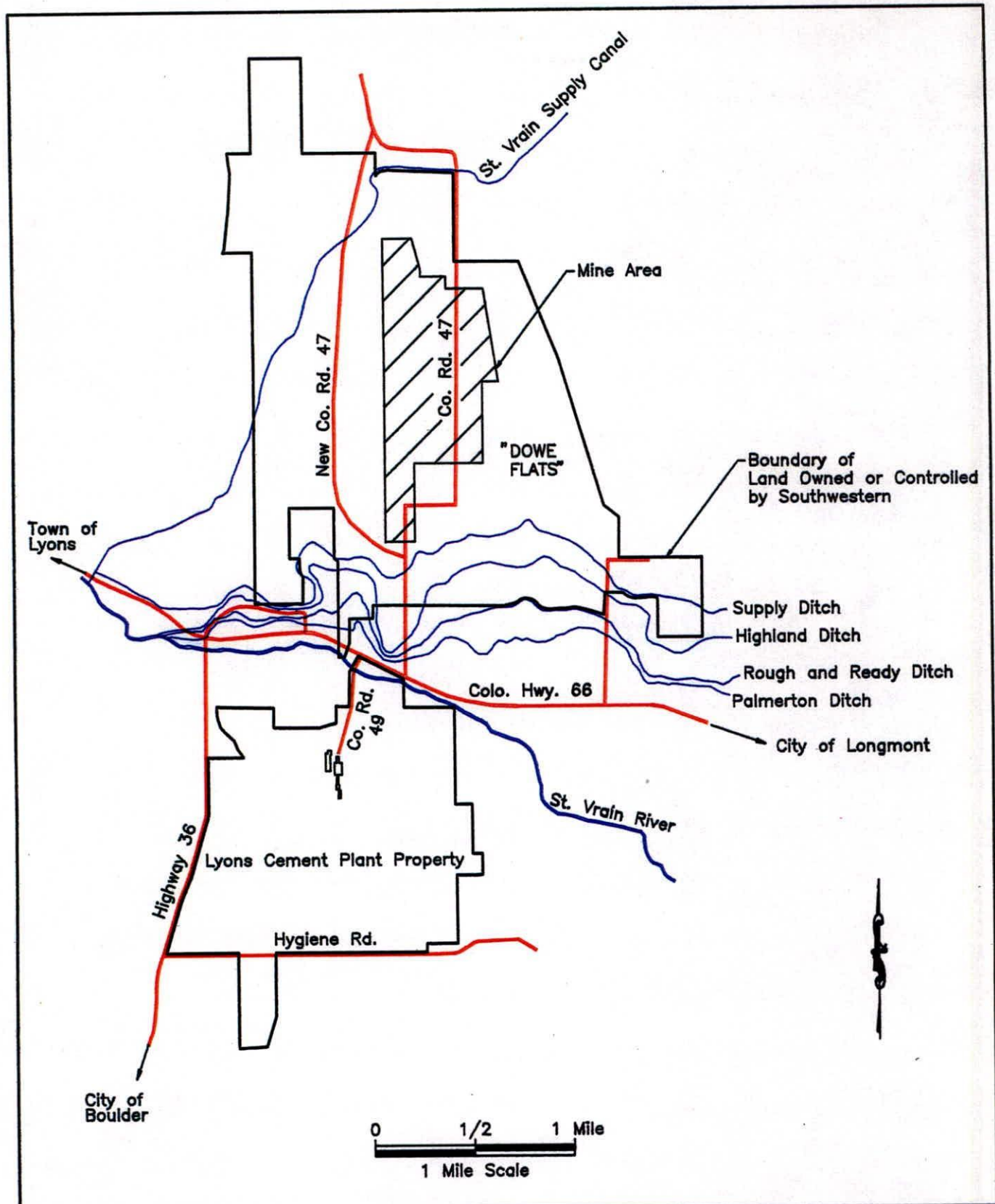
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SHB AGRA, INC.

Engineering & Environmental Services



AGRA

Earth & Environmental Group

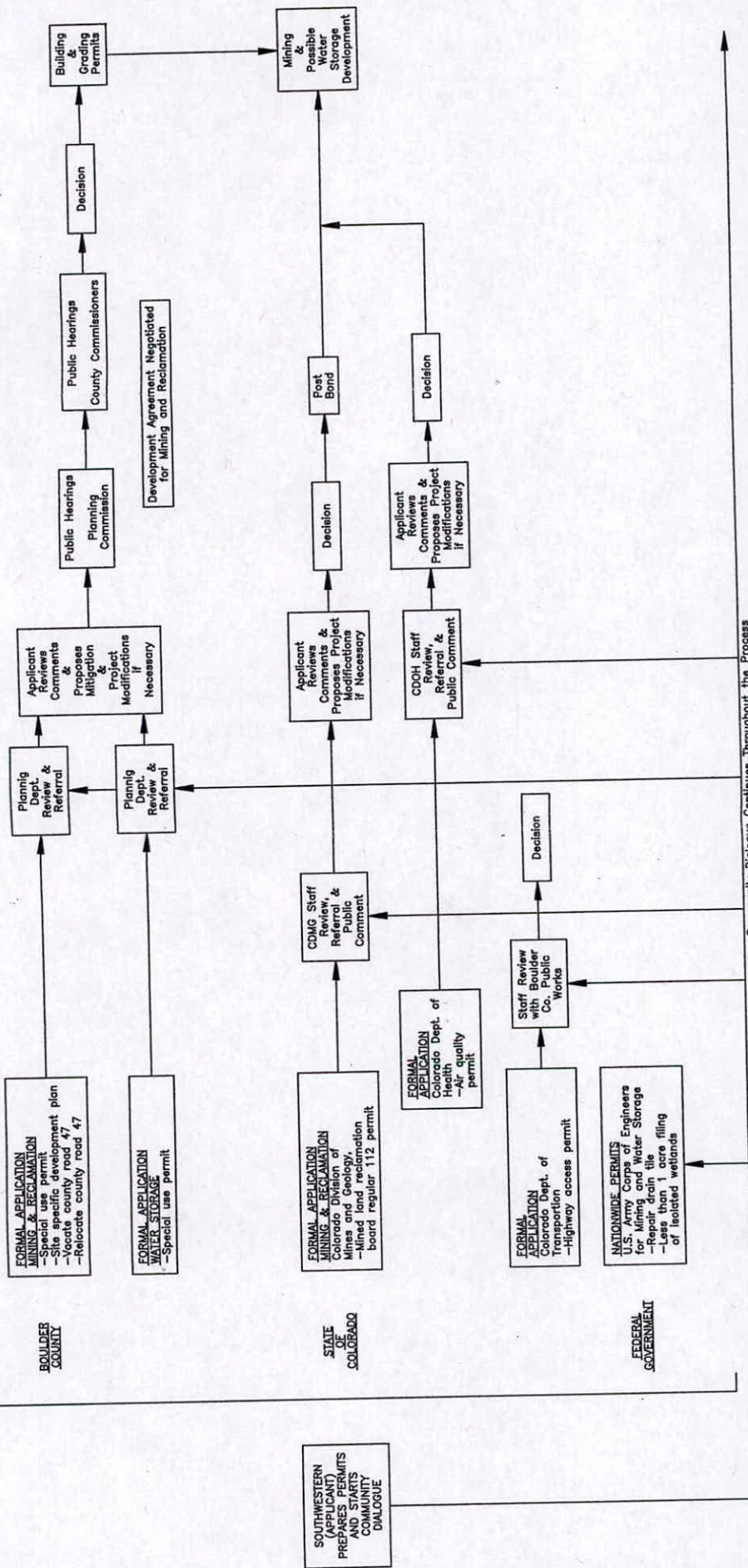
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DOWE FLATS PROJECT
Vicinity Map

Southdown, Inc.
(Southwestern Portland Cement)

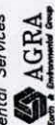
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Time



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SHB AGRA, INC.
Engineering & Environmental Services



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Dowe Flats Project
Public & Agency Review Process
(Southwestern Portland Cement)

Existing Valley



Mining and reclamation complete - 25 years +



Reservoir Built on Reclaimed Surface

